

SMART PARKING PLATFORM

Homework 1: Competitive Research, Vision & Scope,
and Preliminary Software Requirements Specification (SRS)

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CIS 4374 19148

September 10, 2026

Document Status: DRAFT — Living document, updated weekly through end of semester

1. Competitive Landscape and Market Research

1.1 Existing Solutions

Upon further research into the smart-parking space, it becomes apparent that there are two main competitor groups that have minimal overlap between them. Applications such as SpotHero or ParkMobile make it easier for consumers to find and purchase available parking. Alternatively, there are back-office applications, such as T2 Systems and FlashParking, that aid in managing and operating parking lots and garages. There is not currently a company that has a solution that does both of these tasks well, which provides an opportunity for our Smart Parking Platform to implement an integrated system to appeal to consumers as well as business owners/operators of parking facilities.

Product	Strengths	Weaknesses	Differentiation Opportunity
SpotHero (driver app)	Large network, strong pre-booking discounts and guaranteed spots, good event coverage.	Inventory is almost entirely commercial garages/lots, no on-street or municipal coverage, limited real-time occupancy for drive-up spots.	Blend guaranteed pre-booked garage inventory with live on-street availability in one map.
ParkMobile (driver app)	Largest U.S. footprint for on-street meter and municipal lot payment, simple pay-by-zone flow, deep city integrations.	Weak discovery, assumes the driver already found a spot, per-transaction convenience fee, minimal garage reservation capability.	Combine 'find a spot' and 'pay for a spot' so drivers never need a second app.
T2 Systems (operator platform)	Enterprise-grade permit, enforcement, and multi-location reporting, handles complex campus/city ordinance rules, and there's LPR integration.	Built for large operators (universities, cities), heavy implementation, not aimed at small/mid-size lots, no consumer app of its own.	Give small and mid-size operators the same analytics data without an enterprise sales cycle.
FlashParking (operator platform)	Cloud-native PARCS hardware, strong valet workflows, modern dashboard, positioned as the challenger to legacy vendors.	Still primarily a hardware plus back-office play, relies on partner apps for the driver-facing discovery experience.	Ship the driver app and the operator dashboard as one product and not two different integrated ones.

Table 1: Sample competitors: strengths, weaknesses, and differentiation opportunities.

1.2 Summary: How the Smart Parking Platform Will Differ

Unlike current options that separate consumer and owner experiences, the proposed Smart Parking Platform integrates reservations, flexible payments, and real-time space availability for private, on-street, and public parking in one ecosystem. Drivers get an intuitive, feature-rich app; owners get streamlined, actionable tools. This integrated, real-time approach sets it apart in a crowded but fragmented market.

2. Vision and Scope

2.1 How This Project Was Acquired

A metropolitan transportation authority, representing both mid-size downtown parking operators and transportation departments in the city, made a request to WE ARE Software Corp. for development of a Smart Parking Platform to improve parking in their district. WE ARE Software Corp. will develop a custom application for this project.

Requirements will be fleshed out more concretely throughout the project, including market research, interviewing the client for their needs, etc. The timeline and deliverables are fixed for this engagement, but budget is assumed to be unlimited (see Section 2.2.2 for more information on this assumption).

2.2 Company Introduction

WE ARE Software Corp. is a fictitious company created for this project. The company is a software development studio that builds bespoke, custom products for its clients.

2.3 Project Overview

The Smart Parking Platform is a web and mobile system that helps drivers locate, reserve, and pay for parking spaces in real time, while giving parking operators the tools to monitor occupancy, manage pricing, and improve utilization of their facilities. At a high level, the platform must support:

- User registration and authentication
- Real-time parking space availability
- An interactive map of available parking locations
- Parking reservation capability
- Digital payment processing
- Reservation history and receipts
- An administrative dashboard for parking operators
- Occupancy reporting and analytics
- Notifications and alerts for users
- Integration with external mapping and navigation services

2.4 Vision Statement

WE ARE Software Corp. will develop the Smart Parking Platform that will provide operators with web-based tools to manage their lots and drivers with web- and mobile-based tools to find, reserve, and pay for parking in real-time. Our solution will improve parking utilization and reduce wasted time and resources for drivers by providing easy-to-use tools backed by a robust and reliable system. WE ARE Software Corp. will exceed expectations through innovative approaches to the problems of parking that other point-solutions (for example, SpotHero, ParkMobile, T2 Systems, etc.) have not fully realized.

2.5 Scope

2.5.1 In Scope (Semester / Release 1)

- Driver mobile and web applications: registration, search, map view, reservation, payment, history/receipts, notifications.
- Operator administrative dashboard: facility setup, pricing rules, occupancy monitoring, reporting.
- Backend/API layer coordinating availability, reservations, and payments across clients.
- Integration with a third-party payment provider and a third-party mapping/navigation service.

- Compliance groundwork for data privacy and applicable city parking ordinances.

2.5.2 Out of Scope (Future Releases)

- Physical IoT sensor hardware installation (the platform will assume a data feed from operator-side sensors/LPR, not manufacture it).
- Dynamic, demand-based pricing automation (manual pricing rules only in Release 1).
- Valet and EV-charging-specific workflows.
- Multi-city expansion beyond the initial pilot district.

3. Preliminary Software Requirements Specification (SRS) Draft

3.1 Primary Actors

- Driver: an end user searching for, reserving, and paying for parking.
- Parking Operator: staff managing a specific facility's listing, pricing, and reports.
- System Administrator: manages platform-wide users, roles, and integrations.
- Finance / Billing Personnel: handles refunds, disputes, and reconciliation.
- Payment Provider (external): processes digital payments on the platform's behalf.
- Mapping / Navigation Service (external): supplies routing and turn-by-turn guidance.

3.2 Use Case Summary

ID	Use Case	Primary Actor	Brief Description
UC-01	Register Account	Driver	A new driver creates an account with email/phone and basic profile information.
UC-02	Log In / Authenticate	Driver / Operator / Admin	Any registered user securely signs in to access role-appropriate features.
UC-03	Search Nearby Parking	Driver	The driver searches for parking near a destination or current location.
UC-04	View Real-Time Availability on Map	Driver	The driver views an interactive map showing live open/occupied spaces.
UC-05	Reserve Parking Space	Driver	The driver selects and holds a specific space for a chosen arrival window.
UC-06	Cancel Reservation	Driver	Driver cancels an upcoming reservation, subject to the facility's policy.
UC-07	Process Digital Payment	Driver / Payment Provider	The system charges the driver's saved payment method for a reservation or session.
UC-08	View Reservation History & Receipts	Driver	Driver reviews past reservations and downloads/emails a receipt.
UC-09	Navigate to Parking Facility	Driver / Mapping Service	System hands off turn-by-turn navigation to the reserved facility.
UC-10	Receive Notifications & Alerts	Driver	The driver receives alerts for reservation confirmation, expiration, or changes.
UC-11	Extend Active Parking Session	Driver	Driver extends time remaining on an active session before it lapses.
UC-12	Manage Parking Facility Listing	Parking Operator	The operator creates/edits facility details: address, capacity, hours, rules.
UC-13	Set / Update Pricing Rules	Parking Operator	The operator configures base rates and time-based pricing for a facility.
UC-14	Monitor Real-Time Occupancy	Parking Operator	The operator views a live occupancy view of their facility from the dashboard.
UC-15	Generate Occupancy & Revenue Reports	Parking Operator	Operator exports utilization and revenue reports over a selected date range.

Table 2: Preliminary use case set.